



Client Requirement Document

Project

Smart Customer Support Triage Agent

Company

TechEase Cloud

TechEase Cloud provides:

- Cloud storage
- Team collaboration
- Subscription plans
- API platform

We receive around **4,000 support tickets daily**.

Currently, humans manually read every ticket and route it.

This is slow.

We want an AI workflow built using LangGraph.

Business Goal

When a customer submits a support ticket, the AI should:

1. Understand the issue.
2. Categorize it.
3. Decide priority.
4. Look for answers in our knowledge base.
5. Draft a response.
6. Decide whether it can reply automatically or send the ticket to a human.

Input

The API receives JSON like this:

```
{  
  "ticket_id": "TK-10283",  
  "customer_name": "John Smith",  
  "customer_email": "john@email.com",  
  "subscription": "Pro",  
  "message": "I upgraded yesterday but I still cannot access premium features."  
}
```

Ticket Categories

Every ticket must belong to ONE category.

Possible categories:

- Billing
 - Technical
 - Account
 - Feature Request
 - Bug Report
 - Refund
 - General Inquiry
-

Priority

The AI should assign

- Low
- Medium
- High
- Critical

Examples

Critical

- Data loss
- Security breach
- Payment charged multiple times

High

- Cannot login
- API broken
- Premium not activated

Medium

- Bug
- Feature not working

Low

- General questions
 - Feature requests
-

Knowledge Base

We already have documentation.

Examples

Reset Password

Refund Policy

Premium Activation

Billing FAQ

API Rate Limits

Workspace Management

Storage Limits

Authentication Errors

Subscription Upgrade

Invoice Download

The AI should search these documents before generating any answer.

Auto Reply Rules

The AI should automatically reply ONLY IF

- Confidence is high
- A relevant document is found
- The issue is common

Examples



Password reset

Invoice download

Subscription activation

Storage limit

Billing FAQ

Otherwise

Send to Human Support.

Human Escalation Rules

Always escalate when

Customer is angry

Example

I've contacted you five times already.

Legal issue

I'm reporting this to my lawyer.

Refund larger than \$500

Security

Someone hacked my account.

Possible outage

Multiple services unavailable

Low confidence

Draft Response

The AI should generate a polite response.

Example

Hello John,

Thanks for contacting TechEase.

Based on your issue, it appears your premium upgrade has not yet synchronized.

Please try logging out and logging back in.

If the issue remains after 15 minutes, reply to this email and we'll investigate further.

Regards,
Support Team

Output JSON

We expect something similar to

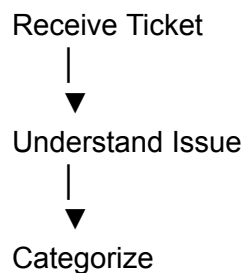
```
{  
  "ticket_id": "TK-10283",  
  "category": "Billing",  
  "priority": "High",  
  "confidence": 0.94,  
  "kb_article": "Premium Activation",  
  "reply": "...",  
  "route": "AUTO_REPLY"  
}
```

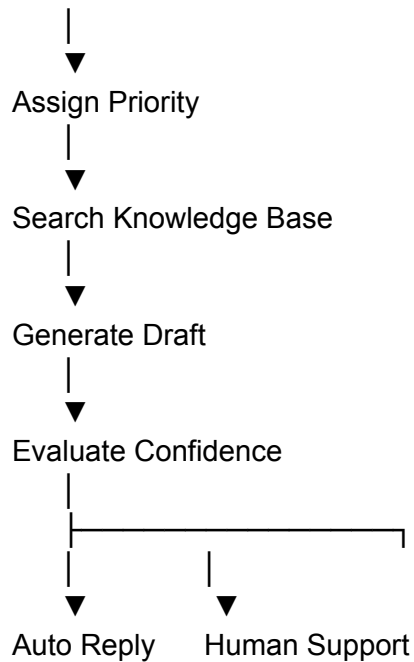
or

```
{  
  "ticket_id": "TK-20912",  
  "category": "Security",  
  "priority": "Critical",  
  "route": "HUMAN_SUPPORT",  
  "reason": "Possible account compromise."  
}
```

Expected Workflow (Business View)

We don't care how it's implemented, but we imagine something like:





Constraints

- Average processing time should be under **5 seconds** per ticket.
 - Every AI decision should be traceable so our support managers can understand why a ticket was routed a certain way.
 - If the AI is uncertain, we prefer escalation over an incorrect automated response.
 - We expect to evaluate and improve the system over time.
-

Out of Scope (Version 1)

Do **not** implement:

- Voice support
 - Live chat
 - Multiple languages
 - Sending emails (just generate the draft)
 - CRM integration
 - Authentication
 - Agentic web search
-

Your Deliverables

As the engineer, you should now decide and implement:

- The LangGraph state schema.
- The nodes in the graph.
- The conditional edges.
- Whether to use tools.
- Whether to use structured outputs.
- How to implement RAG.
- How to calculate or infer confidence.
- How to handle retries or failures.
- What to log and trace in LangSmith.
- Any guardrails or validation you think are necessary.

One final constraint from the client: **we have a budget for only one LLM API call wherever possible.** If you think multiple calls are necessary, be prepared to justify why the additional accuracy or reliability is worth the cost. That trade-off is intentionally left for you to solve, just like in a real client engagement.